



KENYATTA UNIVERSITY

SCHOOL OF HEALTH SCIENCE

DEPARTMENT OF PHARMACEUTICS AND INDUSTRIAL PHARMACY

COURSE CODE AND TITLE: PPA 532 PHARMACEUTICAL MICROBIOLOGY

CONTINUOUS ASSESSMENT TEST 2

TUESDAY 15TH JULY 2025

11:30AM – 12:30PM

ANSWER ALL QUESTIONS. 30MARKS

1. Which microorganism is commonly used in the assay of penicillin?

A. *Bacillus subtilis*

B. *Escherichia coli*

☒ C. *Staphylococcus aureus* -

D. *Pseudomonas aeruginosa*

2. Which method is commonly used in microbiological assay?

A. Gravimetric

☒ B. Cylinder plate -

C. Titrimetric

D. Spectrophotometric

3. Which is not an advantage of microbiological assay?

A. Measures biological activity

B. Simple and inexpensive

C. Highly specific

☒ D. Not time-consuming

4. Which organism is used in the assay of vitamin B12?

- A. *Escherichia coli*
- ☒ B. *Lactobacillus leichmannii*
- C. *Bacillus subtilis*
- D. *Aspergillus niger* -

5. The endpoint in a microbiological assay can be determined by:

- A. Color change
- ☒ B. Turbidity -
- C. Weight loss
- D. Gas evolution

6. Microbiological assay is useful for:

- A. Measuring physical properties
- B. Analyzing inorganic compounds
- ☒ C. Determining potency of antibiotics -
- D. Determining melting point

7. Which medium is used for microbiological assay of antibiotics?

- A. Nutrient broth
- B. Sabouraud agar
- ☒ C. Mueller-Hinton agar
- D. EMB agar

8. Which antibiotic can be assayed using *Bacillus pumilus*?

A. Tetracycline

B. Chloramphenicol

C. Polymyxin B

☒ D. Neomycin

9. In cylinder plate method, the antibiotic diffuses into:

☒ A. Solid agar

B. Liquid broth

C. Gas phase

D. None

10. Which assay method uses spectrophotometer?

☒ A. Turbidimetric

B. Cylinder plate

C. Cup plate

D. Agar diffusion

11. The standard curve in microbiological assay is plotted as:

☒ A. Zone diameter vs log concentration

B. Absorbance vs time

C. pH vs temperature

D. Colony count vs time

12. Which of the following method can be used to determine the number of bacteria quantitatively?
- A. Spread-plate
 - B. Streak-plate
 - ☒ C. Pour-plate and spread plate
 - D. Pour plate
13. Which of the following microorganism produces dextran?
- ☒ A. *Leuconostoc mesenteroides*
 - B. *Streptomyces olivaceus*
 - C. *Bacillus thuringiensis*
 - D. *Bacillus polymyxa*
14. The strain of fungi used for the large scale production of penicillin is;
- A. *Penicillium chrysogenum*
 - B. *P-notatum*
 - C. *Streptomyces Aureus*
 - D. *Saccharomyces* sps
15. *Aspergillus niger* is used in the industrial production of;
- A. Ethanol
 - B. Penicillin
 - C. Citric acid
 - D. Lactic acid
16. Which bacteria commonly produce beta-lactamase?
- A. *Streptococcus pyogenes*
 - B. *Escherichia coli*
 - C. *Mycobacterium tuberculosis*
 - D. *Clostridium tetani*

17. Which resistance mechanism is associated with tetracyclines?

- A. Target modification
- B. Efflux pump
- C. Enzymatic inactivation
- D. Biofilm formation

18. The *mecA* gene is responsible for resistance in:

- A. MRSA
- B. VRE
- C. ESBL
- D. CRE

19. Which enzyme provides resistance to aminoglycosides?

- A. Beta-lactamase
- B. Acetyltransferase
- C. Carbapenemase
- D. DNA gyrase

20. Which is an example of intrinsic resistance?

- A. MRSA producing PBP2a
- B. *E. coli* producing ESBL
- C. *Pseudomonas* resistance to penicillin
- D. *Klebsiella* with KPC

21. Which mechanism is used by bacteria to resist macrolides?

- A. Beta-lactamase production
- B. Methylation of ribosomal RNA
- C. Carbapenemase production
- D. Cell wall thickening

22. Which is an example of acquired resistance?

- A. *Pseudomonas* resistance to penicillin
- B. *Mycoplasma* resistance to beta-lactams
- C. MRSA
- D. *Enterococcus* intrinsic resistance

23. Which antibiotic resistance mechanism is horizontally transferred?

- A. Target site mutation
- B. Biofilm formation
- C. Enzymatic inactivation via plasmid
- D. Intrinsic permeability barrier

24. The by-product during streptomycin production is;

- A. Vitamin A
- B. Proline
- C. Vitamin B12
- D. Citric acid

25. The best medium for the production of Penicillin is;

- A. Nutrient agar
- B. Corn steep liquor
- C. Sulfite waste liquor
- D. Whey

26. What does the survivor curve plot?

- A. Temperature vs time
- B. Microbial count vs temperature
- C. Log survivors vs time
- D. Sterilant concentration vs time

27. The higher the D-value:

- A. The faster the sterilization
- B. The more resistant the organism
- C. The less heat required
- D. The higher the F0

28. What does a higher Z-value indicate?

- A. Microbe is more heat sensitive
- B. Microbe is more heat resistant
- C. D-value is lower
- D. F0 is constant

29. If initial count is 1,000,000 CFU and D-value is 1 min, how long to reduce to 1 CFU?

- A. 5 min
- B. 6 min
- C. 10 min
- D. 1 min

30. The Arrhenius equation relates the rate constant to:

A. Temperature

B. Pressure

C. Concentration

☒ D. Time